Three variables

Keep track of three values and the signs between terms.

Name: _____ Suggested time: 10 minutes

Show your working. Use brackets when substituting negative values.

01 Given a=3, b=3, c=1

Evaluate: 3a - 6b +2c

02 Given a=0, b=0, c=0

Evaluate: 5a - 4b +2c

03 Given a=0, b=-1, c=-2

Evaluate: 4a - 4b +4c

04 Given a=3, b=-4, c=-3

Evaluate: 6a - 3b +5c

05 Given a=3, b=0, c=-5

Evaluate: 4a - 3b +5c

06 Given a=1, b=3, c=-3

Evaluate: 4a - 3b +5c

Answer sheet

Three variables | 6 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 -7

02 0

03 -4

04 15

05 -13

06 -20

Worked solutions

Three variables

01 Given a=3, b=3, c=1

$3a - 6b + 2c$

Substitute values.

$3\times 3 - 6\times 3 + 2\times 1$

$9 - 18 + 2$

-7

02 Given a=0, b=0, c=0

$5a - 4b + 2c$

Substitute values.

$5\times 0 - 4\times 0 + 2\times 0$

0

0

03 Given a=0, b=-1, c=-2

$4a - 4b + 4c$

Substitute values.

$4\times 0 - 4\times (-1) + 4\times (-2)$

$0 + 4 - 8$

-4

04 Given a=3, b=-4, c=-3

$6a - 3b + 5c$

Substitute values.

$6\times 3 - 3\times (-4) + 5\times (-3)$

$18 + 12 - 15$

15

Worked solutions

Three variables

05 Given a=3, b=0, c=-5

$4a - 3b + 5c$

Substitute values.

$4\times 3 - 3\times 0 + 5\times (-5)$

$12 - 25$

-13

06 Given a=1, b=3, c=-3

$4a - 3b + 5c$

Substitute values.

$4\times 1 - 3\times 3 + 5\times (-3)$

$4 - 9 - 15$

-20